Trading Platform

2025

Fullstack Development for a Futuristic Quant Platform
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1 AML Simulation and portfolio optimization

- Kopeliovich, Yaacov & Pokojovy, Michael (2024). “Portfolio Optimization with Feedback Strategies Based on Artificial Neural Networks.” *Finance Research Letters*, Vol. 69.
- Bates, David S. (2006). “Maximum Likelihood Estimation of Latent Affine Processes.” *The Review of Financial Studies*, Vol. 19, No. 3 (Autumn), pp. 909-965.
- Marco Rando, Cesare Molinari, Lorenzo Rosasco, and Silvia Villa. *A Structured Tour of Optimization with Finite Differences.* arXiv preprint arXiv:2505.19720, 2025. https://arxiv.org/abs/2505.1972

1.1 Modeling and simulating Underlying asset dynamics:

1.1.1 Bates AML for SVj1

$$dS_t = \Big[\mu_0 + (\mu_1 - \tfrac{1}{2})\,V_t - (\lambda_0 + \lambda_1 V_t)\,\bar{k}\Big]\,dt + \sqrt{V_t}\Big(\rho\,dW_{1t} + \sqrt{1-\rho^2}\,dW_{2t}\Big) + \gamma_s\,dN_t, \tag{1}$$

$$dV_t = (\alpha - \beta V_t)\,dt + \sigma\sqrt{V_t}\,dW_{1t}, \tag{2}$$

(W_{1t}, W_{2t}) independent Brownian motions$(dW_{1t}, dW_{2t}) = 0)$,
inst. corr. between return and variance shocks is ρ via the dW_{1t} term,
N_t is a Poisson counter with intensity $\lambda_0 + \lambda_1 V_t$,
$\gamma_s \sim \mathcal{N}(\gamma, \delta^2)$ (log-jump size), $\qquad \bar{k} \equiv e^{\gamma + \frac{1}{2}\delta^2} - 1.$

$$\arg\max_{\theta} T(\theta) \tag{3}$$

$$T(\theta) = -loglikelihood \tag{4}$$

$$T(\theta) = -\sum_{i=1}^{N} p(y_{t+1} y_t|\theta) \tag{5}$$

$$\tag{6}$$

Algorithm 1 Parameter Optimization Loop

$\theta_{\text{init}} \leftarrow (\mu_0, \mu_1, \rho, \alpha, \beta, \sigma, \lambda_0, \lambda_1, k, dt)$
$T_{\text{threshold}} \leftarrow 0.1$
Initialize inputs
{N: Evaluation Budget count, h: gradient descent step size,prior$_\theta$ $\leftarrow$ initial prior params}
$\theta_{\text{old}} = \theta_{\text{init}}$
j = 0
while $T(\theta) > T_{\text{threshold}} \cap i < N$ **do**
    Update prior$_{\theta_{\text{init}}}$
    $B = \{s_1..s_{number\ of\ batches}\}$(Sample start Idxs of y)
    s_{len} : Sample Length
    $P = [p^{(1)}, \ldots, p^{(\ell)}]$: Choose structured direction matrix
    $g \leftarrow 0$: Intialize gradient estimate
    Compute perturbed log-likelihood:
    for $i = 1$ to ℓ **do**
        $\theta_{\text{new}} \leftarrow \theta_{old} + hp^{(i)}$
        for $\theta \in \{\theta_{old}, \theta_{new}\}$ **do**
            $LL_{\theta,total} = 0$
            for $\{y_0, \ldots, y_T\} \in \{\{y_{ind:(y_{ind}+s_{len})}\}\} s.t.\ ind \in B$ **do**
                $LogLikelihood_{ind} = 0$
                for $y_t \in \{y_0, \ldots, y_T\}$ **do**
                    Store: y_{t-1}
                    $cf : F(\theta, \phi, \psi = 0) = e^{C(\phi,0)} G_{prior_\theta}[D(\phi, 0)]$
                    $p(y_{t+1} \mid Y_t) \approx \frac{\Delta\Phi}{2\pi} \sum_{k=-K}^{K} cf\, e^{-i\Phi_k y_{t+1}}, \qquad \Phi_k = k\,\Delta\Phi.$
                    $LogLikelihood_{ind} += log(p(y_{t+1} \mid Y_t))$
                end for
                $LL_{\theta,Total} \leftarrow LL_{\theta,Total} + LogLikelihood_{ind}$
            end for
            $LL_{Cache} \leftarrow LL_{\theta,Total}$
        end for
        Update Gradient:
        $L\bar{L}_{new} \in LL_{Cache}$
        $L\bar{L}_{old} \in LL_{Cache}$
        $$g \leftarrow g + \frac{(L\bar{L}_{new} - L\bar{L}_{old})}{h}\, p^{(i)}$$
    end for
    Choose step size η satisfying the stochastic Armijo condition:
    $T(\theta_{\text{old}} - \eta g) \leq T(\theta_{\text{old}}) - \ell_1 \eta \|g\|^2 + 2\,\widehat{Var}[\epsilon]$
    $\theta_{old} \leftarrow \theta_{\text{old}} - \eta g$
    $j = j + 1$
end while

1.1.2 Utility Objective

1.1.3 Simulate ealth dyaics under feedback allocatio strategies

1.1.4 Training the ANN to optimize the expected utility of the underlying wealth process